GUIDELINE DOCUMENT

LAB - CCNA 2 v7.2

Physical LAB

(HSRP Configuration)

Designed by	:	Praing Senghong
Mobile	:	086 737 377 | 090 737 377
Email	:	praingsenghong@gmail.com
Linkedin	:	https://linkedin.com/in/praingsenghong
Telegram	:	https://t.me/senghongpraing

Version 1.0 | July 2025

Contents

Table Contents

Figure Contents

1 Introduction

In networking, ensuring **high availability and redundancy** is critical to maintaining uninterrupted access to network resources. **First Hop Redundancy Protocol (FHRP)** is a category of protocols designed to eliminate single points of failure associated with the default gateway in IP networks. Without FHRP, if the default gateway fails, end hosts lose access to external networks.

FHRP protocols allow multiple routers to collaboratively share a virtual IP address, which serves as a highly available default gateway. If the primary router or gateway becomes unavailable, a backup router automatically takes over responsibility, ensuring continuous network connectivity.

The most common FHRP protocols include:

- **Hot Standby Router Protocol (HSRP):** Cisco proprietary protocol widely used in enterprise environments.
- **Virtual Router Redundancy Protocol (VRRP):** Industry-standard, open protocol with functionality similar to HSRP.
- **Gateway Load Balancing Protocol (GLBP):** Cisco proprietary protocol providing redundancy along with load balancing capabilities among multiple routers.

This guideline we selected the **HSRP** to do the lab.

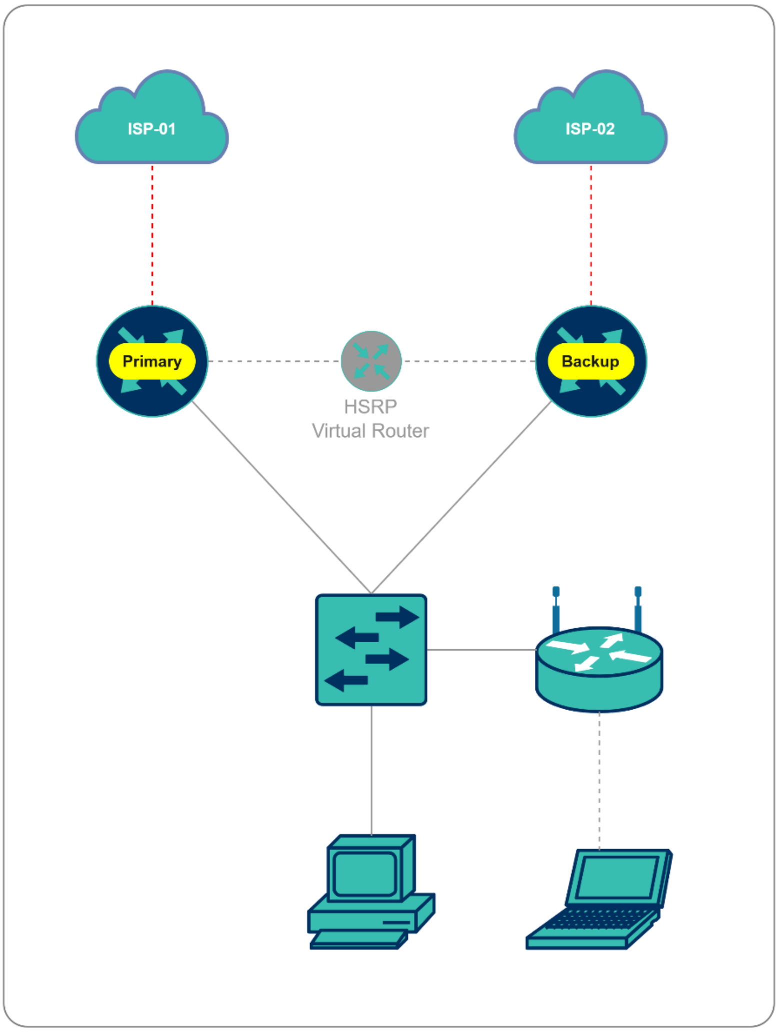

Figure 1 - General Architecture HSRP

2 Preparation and Planning

2.1 Design Lab Topology

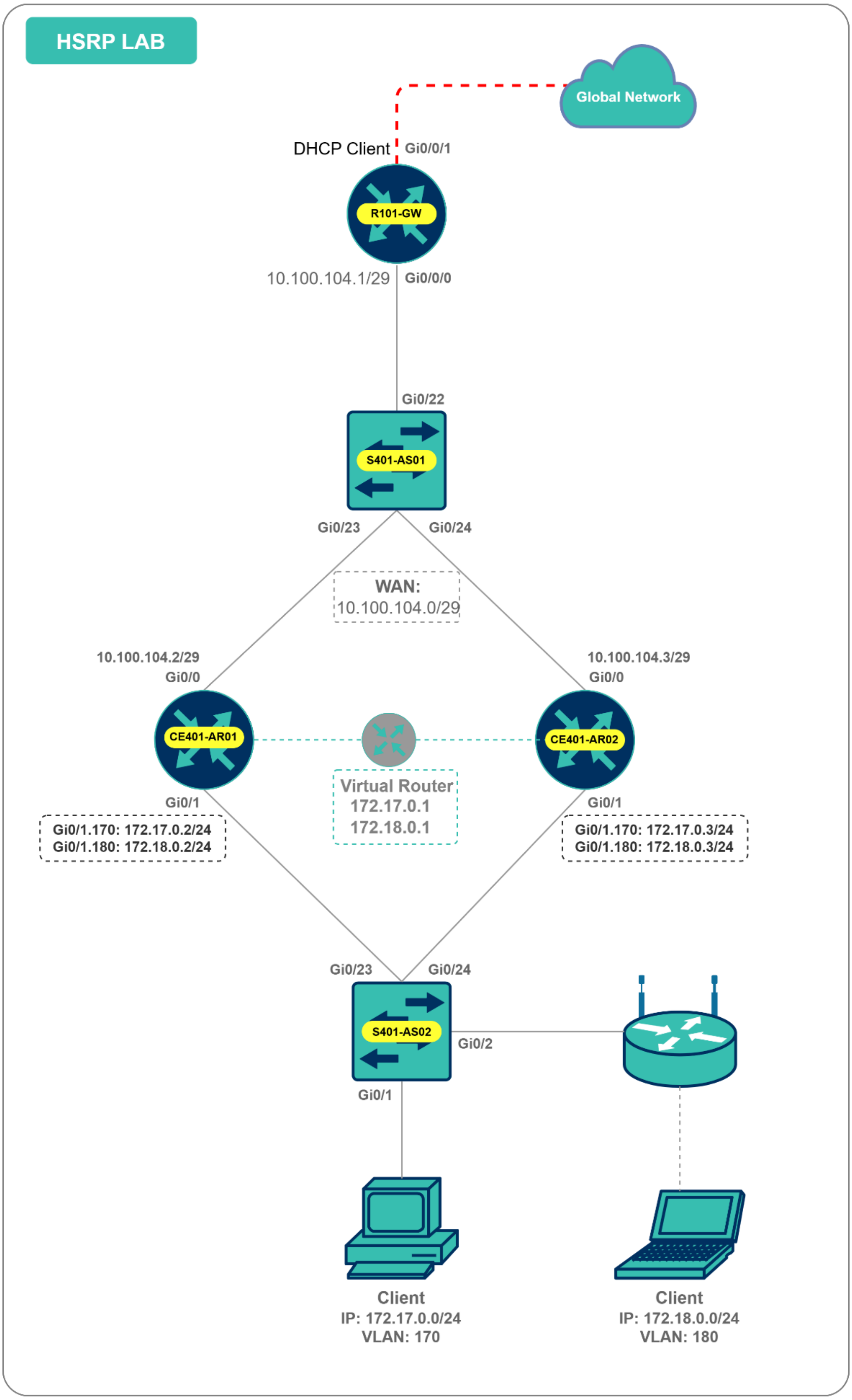

Figure 2 - Lab Topology

2.2 IP Addressing Plan

Table 1 - Subnet Planning

Purpose	Subnet	Subnet Mask	Remarks
Internet Link	10.100.104.0	255.255.255.248	WAN Connection
Client VLAN 170	172.17.0.0	255.255.255.0	Client subnet
Client VLAN 180	172.18.0.0	255.255.255.0	Client subnet
Loopback0	10.1.254.0	255.255.255.0	This subnet Loopback0 and Router ID

Table 2 - Device Interface IP Assignments

Device Name	Interface	VLAN/Subnet	IP Address	Subnet Mask	Remarks
R101-GW	Gi0/0/1				DHCP Client
	Gi0/0/0		10.100.104.1	255.255.255.248	Gateway router
CE401-AR01	Gi0/0		10.100.104.2	255.255.255.248	Wan interface
	Gi0/1.170	VLAN170	172.17.0.2	255.255.255.0	HSRP Active Router
	Gi0/1.180	VLAN180	172.18.0.2	255.255.255.0	HSRP Standby Router
	L0	Loopback0	10.1.254.1	255.255.255.255	Loopback0 and Router ID
CE401-AR02	Gi0/0		10.100.104.3	255.255.255.248	Wan interface
	Gi0/1.170	VLAN 170	172.17.0.3	255.255.255.0	HSRP Standby Router
	Gi0/1.180	VLAN 180	172.18.0.3	255.255.255.0	HSRP Active Router
	L0	Loopback0	10.1.254.2	255.255.255.255	Loopback0 and Router ID
HSRP VIP	VLAN170	VLAN170	172.17.0.1	255.255.255.0	Virtual Gateway for clients
HSRP VIP	VLAN180	VLAN170	172.18.0.1	255.255.255.0	Virtual Gateway for clients
S401-AS02	VLAN170	VLAN170	172.17.0.5	255.255.255.0	IP Management

3 Configuration

3.1 Basic Device Configuration

Table 3 - Basic Device Configuration

Device	Step	Command	Purpose
S401-AS02	Step 1	Switch> Switch>enable Switch# configure terminal	Enters global configuration mode.
	Step 2	Switch(config)#hostname S401-AS02	Specifies the name for the switch.
	Step 3	S401-AS02(config)#enable secret Cisco@dmin123	Specifies an encrypted password to prevent unauthorized access to the switch.
	Step 4	S401-AS02(config)#username cisco secret Cisco@dmin123	Create username and password.
	Step 5	S401-AS02(config)#banner motd #Unauthorized access is strictly prohibited!#	Banner MOTD
	Step 6	S401-AS02(config)#ip domain-name cisco.com S401-AS02(config)#ip name-server 8.8.8.8	
	Step 7	S401-AS02(config)#crypto key generate rsa general-keys modulus 1024	generate RSA keys for SSH access
	Step 8	S401-AS02(config)#line console 0 S401-AS02(config-line)#login local S401-AS02(config-line)#exec-timeout 3	Configure line console
	Step 9	S401-AS02(config)#line vty 0 4 S401-AS02(config-line)#transport input ssh S401-AS02(config-line)#login local S401-AS02(config-line)#exec-timeout 3	Configure line vty to allow SSH only.
	Step 10	S401-AS02(config)#vlan 170 S401-AS02(config-vlan)#exit S401-AS02(config)#interface vlan 170 S401-AS02(config-if)#ip address 172.17.0.5 255.255.255.0 S401-AS02(config-if)#exit S401-AS02(config)#ip default-gateway 172.17.0.1	Configure IP management on the switch.

	Step 1	Router> Router>enable Router#configure terminal	Enters global configuration mode.
	Step 2	Router(config)#hostname CE401-AR01	Specifies the name for the router.
	Step 3	CE401-AR01(config)#enable secret Cisco@dmin123	Specifies an encrypted password to prevent unauthorized access to the router.
	Step 4	CE401-AR01(config)#username cisco secret Cisco@dmin123	Create username and password.
	Step 5	CE401-AR01(config)#banner motd #Unauthorized access is strictly prohibited!#	Banner MOTD
CE401-AR01	Step 6	CE401-AR01(config)#ip domain-name cisco.com CE401-AR01(config)#ip name-server 8.8.8.8	
	Step 7	CE401-AR01(config)#crypto key generate rsa general-keys modulus 1024	generate RSA keys for SSH access
	Step 8	CE401-AR01(config)#line console 0 CE401-AR01(config-line)#login local CE401-AR01(config-line)#exec-timeout 3	Configure line console
	Step 9	CE401-AR01(config)#line vty 0 4 CE401-AR01(config-line)#transport input ssh CE401-AR01(config-line)#login local CE401-AR01(config-line)#exec-timeout 3	Configure line vty to allow SSH only.
	Step 10	CE401-AR01(config)#interface loopback 0 CE401-AR01(config-if)#ip address 10.1.254.1 255.255.255.255	Configure IP management on the router.
	Step 1	Router> Router>enable Router#configure terminal	Enters global configuration mode.
	Step 2	Router(config)#hostname CE401-AR02	Specifies the name for the router.
	Step 3	CE401-AR02(config)#enable secret Cisco@dmin123	Specifies an encrypted password to prevent unauthorized access to the router.
	Step 4	CE401-AR02(config)#username cisco secret Cisco@dmin123	Create username and password.
	Step 5	CE401-AR02(config)#banner motd #Unauthorized access is strictly prohibited!#	Banner MOTD
CE401-AR02	Step 6	CE401-AR02(config)#ip domain-name cisco.com CE401-AR02(config)#ip name-server 8.8.8.8	
	Step 7	CE401-AR02(config)#crypto key generate rsa general-keys modulus 1024	generate RSA keys for SSH access
	Step 8	CE401-AR02(config)#line console 0 CE401-AR02(config-line)#login local CE401-AR02(config-line)#exec-timeout 3	Configure line console
	Step 9	CE401-AR02(config)#line vty 0 4 CE401-AR02(config-line)#transport input ssh CE401-AR02(config-line)#login local CE401-AR02(config-line)#exec-timeout 3	Configure line vty to allow SSH only.
	Step 10	CE401-AR02(config)#interface loopback 0 CE401-AR02(config-if)#ip address 10.1.254.2 255.255.255.255	Configure IP management on the router.

3.2 Create VLAN and Port Assignment

Table 4 - Create VLAN and Port Assignment

Device	Step	Command	Purpose
S401-AS02	Step 1	S401-AS02(config)#vlan 170 S401-AS02(config-vlan)#name Client-VLAN170 S401-AS02(config-vlan)#exit S401-AS02(config)#vlan 180 S401-AS02(config-vlan)#name Client-VLAN180 S401-AS02(config-vlan)#exit	Create VLAN

	Step 2	S401-AS02(config)#interface gigabitEthernet 0/1 S401-AS02(config-if)#switchport mode access S401-AS02(config-if)#switchport access vlan 170 S401-AS02(config-if)#exit S401-AS02(config)#interface gigabitEthernet 0/2 S401-AS02(config-if)#switchport mode access S401-AS02(config-if)#switchport access vlan 180 S401-AS02(config-if)#exit	Assign VLAN to access port.
	Step 3	S401-AS02(config)#interface range gigabitEthernet 0/23-24 S401-AS02(config-if)#switchport mode trunk S401-AS02(config-if)#switchport trunk allowed vlan 170,180 S401-AS02(config-if)#exit	Allowed VLAN on trunk port to router.

3.3 HSRP Configuration

Table 5 - HSRP Configuration

Device	Step	Command	Purpose
CE401-AR01	Step 1	CE401-AR01(config-if)#interface gigabitEthernet 0/1.170 CE401-AR01(config-subif)#encapsulation dot1Q 170 CE401-AR01(config-subif)#ip address 172.17.0.2 255.255.255.0 CE401-AR01(config-subif)#standby 170 ip 172.17.0.1 CE401-AR01(config-subif)#standby 170 priority 110 CE401-AR01(config-subif)#standby 170 preempt CE401-AR01(config-subif)#exit	Configure HSRP active for VLAN 170.
	Step 2	CE401-AR01(config)#interface GigabitEthernet0/1.180 CE401-AR01(config-subif)# encapsulation dot1Q 180 CE401-AR01(config-subif)# ip address 172.18.0.2 255.255.255.0 CE401-AR01(config-subif)# standby 180 ip 172.18.0.1 CE401-AR01(config-subif)# standby 180 priority 90 CE401-AR01(config-subif)# standby 180 preempt CE401-AR01(config-subif)#exit	Configure HSRP standby for VLAN 180.
CE401-AR02	Step 1	CE401-AR02(config)#interface GigabitEthernet0/1.170 CE401-AR02(config-subif)# encapsulation dot1Q 170 CE401-AR02(config-subif)# ip address 172.17.0.3 255.255.255.0 CE401-AR02(config-subif)# standby 170 ip 172.17.0.1 CE401-AR02(config-subif)# standby 170 priority 90 CE401-AR02(config-subif)# standby 170 preempt CE401-AR02(config-subif)#exit	Configure HSRP standby for VLAN 170.
	Step 2	CE401-AR02(config)#interface GigabitEthernet0/1.180 CE401-AR02(config-subif)# encapsulation dot1Q 180 CE401-AR02(config-subif)# ip address 172.18.0.3 255.255.255.0 CE401-AR02(config-subif)# standby 180 ip 172.18.0.1 CE401-AR02(config-subif)# standby 180 priority 110 CE401-AR02(config-subif)# standby 180 preempt CE401-AR02(config-subif)#exit	Configure HSRP active for VLAN 180.

3.4 OSPF Configuration

Table 6 - OSPF Configuration

Device	Step	Command	Purpose
CE401-AR01	Step 1	CE401-AR01(config)#router ospf 1 CE401-AR01(config-router)#router-id 10.1.254.1 CE401-AR01(config-router)#network 10.1.254.1 0.0.0.0 area 0 CE401-AR01(config-router)#network 172.17.0.0 0.0.0.255 area 0 CE401-AR01(config-router)#network 172.18.0.0 0.0.0.255 area 0 CE401-AR01(config-router)#exit	Configure OSPF
CE401-AR02	Step 1	CE401-AR02(config)#router ospf 1 CE401-AR02(config-router)#router-id 10.1.254.2	Configure OSPF

		CE401-AR02(config-router)#network 10.1.254.2 0.0.0.0 area 0 CE401-AR02(config-router)#network 172.17.0.0 0.0.0.255 area 0 CE401-AR02(config-router)#network 172.18.0.0 0.0.0.255 area 0 CE401-AR02(config-router)#exit	

3.5 IP SLA and Static Routing

Table 7 - Configure IP SLA & Static Default Routing

Device	Step	Command	Purpose
CE401-AR01	Step 1	CE401-AR01(config)#ip sla 1 CE401-AR01(config-ip-sla)#icmp-echo 10.100.104.1 source-interface gi0/0 CE401-AR01(config-ip-sla-echo)#frequency 30 CE401-AR01(config-ip-sla-echo)#exit CE401-AR01(config)#ip sla schedule 1 life forever start-time now	IP SLA Configuration for WAN tracking
	Step 2	CE401-AR01(config)#track 1 ip sla 1 reachability CE401-AR01(config-track)#exit	Tracking SLA for static routing
	Step 3	CE401-AR01(config)#ip route 0.0.0.0 0.0.0.0 10.100.104.1 track 1	Static Default route
CE401-AR02	Step 1	CE401-AR02(config)#ip sla 1 CE401-AR02(config-ip-sla)#icmp-echo 10.100.104.1 source-interface gi0/0 CE401-AR02(config-ip-sla-echo)#frequency 30 CE401-AR02(config-ip-sla-echo)#exit CE401-AR02(config)#ip sla schedule 1 life forever start-time now	IP SLA Configuration for WAN tracking
	Step 2	CE401-AR02(config)#track 1 ip sla 1 reachability CE401-AR02(config-track)#exit	Tracking SLA for static routing
	Step 3	CE401-AR02(config)#ip route 0.0.0.0 0.0.0.0 10.100.104.1 track 1	Static Default route

3.6 NAT Configuration

Table 8 - Configure NAT

Device	Step	Command	Purpose
R101-GW	Step 1	R101-GW(config)# interface GigabitEthernet0/0/0 R101-GW(config-if)# ip address 10.100.104.1 255.255.255.248 R101-GW(config-if)# ip nat inside R101-GW(config-if)#no shutdown	NAT on the interface inside
	Step 2	R101-GW(config)# interface GigabitEthernet0/0/1 R101-GW(config-if)# ip address dhcp R101-GW(config-if)# ip nat outside R101-GW(config-if)#no shutdown	NAT on the interface outside
	Step 3	R101-GW(config)# access-list 1 permit 10.100.104.0 0.0.0.7 R101-GW(config)# ip nat inside source list 1 interface Gi0/0/1 overload R101-GW(config)#ip route 0.0.0.0 0.0.0.0 10.199.20.1	Access NAT
CE401-AR01	Step 1	CE401-AR01(config)#interface gigabitEthernet 0/0 CE401-AR01(config-if)#ip address 10.100.104.2 255.255.255.248 CE401-AR01(config-if)#ip nat outside CE401-AR01(config-if)#no shutdown	NAT on the interface outside
	Step 2	CE401-AR01(config)#interface gi0/1.170 CE401-AR01(config-subif)#ip nat inside CE401-AR01(config-if)#no shutdown CE401-AR01(config)#interface gi0/1.180 CE401-AR01(config-subif)#ip nat inside CE401-AR01(config-if)#no shutdown	NAT on the interface inside

	Step 3	CE401-AR01(config)#access-list 10 permit 172.17.0.0 0.0.0.255 CE401-AR01(config)#access-list 20 permit 172.18.0.0 0.0.0.255 CE401-AR01(config)#ip nat inside source list 10 interface Gi0/0 overload CE401-AR01(config)#ip nat inside source list 20 interface Gi0/0 overload	Access NAT
CE401-AR02	Step 1	CE401-AR02(config)#interface gigabitEthernet 0/0 CE401-AR02(config-if)#ip address 10.100.104.3 255.255.255.248 CE401-AR02(config-if)#ip nat outside CE401-AR02(config-if)#no shutdown	NAT on the interface outside
	Step 2	CE401-AR02(config)#interface gi0/1.170 CE401-AR02(config-subif)#ip nat inside CE401-AR02(config-if)#no shutdown CE401-AR02(config)#interface gi0/1.180 CE401-AR02(config-subif)#ip nat inside CE401-AR02(config-if)#no shutdown	NAT on the interface inside
	Step 3	CE401-AR02(config)#access-list 10 permit 172.17.0.0 0.0.0.255 CE401-AR02(config)#access-list 20 permit 172.18.0.0 0.0.0.255 CE401-AR02(config)#ip nat inside source list 10 interface Gi0/0 overload CE401-AR02(config)#ip nat inside source list 20 interface Gi0/0 overload	Access NAT

4 Verification & Testing Commands

4.1 Verify OSPF

show ip ospf neighbor

4.2 Verify HSRP status

show standby brief

4.3 Verify NAT translations

show ip nat translations

4.4 Verify IP SLA

show ip sla statistics

4.5 Verify static route tracking

show track